TT-I-564C
November 11, 1975
SUPERSEDING
TT-I-564B
February 28, 1974

FEDERAL SPECIFICATION

INK, STENCIL, DUPLICATING MACHINE, BLACK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ink for use in the stencil-duplicating process (see 6.1).

1.2 Classification. Duplicating-machine stencil ink covered by this specification shall be of the following types and classes, as specified (see 6.2):

Type I - Deleted (see 6.3).

Type II - Water-in-oil emulsion-base ink.

Type III - Hard-set ink.

Type IV - Fast-drying inks.

Class 1 - Water-base
Class 2 - Oil-base

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-121 - Paper, Bond and Writing, White and Colored.
UU-P-388 - Paper, Mimeograph.
UU-S-1205 - Stencil, Duplicating-Machine.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

FSC 7510

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Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling & Testing.
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and workmanship. The manufacturer is given wide latitude in his selection of materials and processes of manufacture, provided the ink meets the requirements herein. The ink shall be homogeneous and free of foreign matter.

3.2 Shelf life. Type II and type IV shall not skin or cake and shall remix with the vehicle when agitated (see 4.3.7). All inks shall meet the requirements specified here in 1 year from date of delivery (see 4.1.1).

3.3 Color. The color of the ink as applied, shall be jet black or blue black (see 4.2.2.1).

3.4 Physical properties. The inks shall have the properties specified in table I, when tested as specified in 4.3.1, 4.3.2, and 4.3.3.2.

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TABLE I. Physical properties

	Specific gravity at 25°C	Stromer viscoscity seconds Min	Max	Drying time seconds
Type II	1.025	160	200	600
Type III	1.200	160	200	600
Type IV	1.200	130	150	30

3.5 Reproduction requirements. The ink shall make legible copies without strike through, smudge, vehicle stain or offset when the following copies are runoff (see 4.3.3).

Type II - 1000 copies; face and back, 3000 copies face only.
Type III - 2000 copies; face only.
Type IV - 1000 copies; face and back.

The first 100 copies shall remain legible after 48 hours when tested as specified in 4.3.3.1. The stencil shall not fill in or become tacky after 24 hours when tested as specified in 4.3.4. The ink shall not build up under the stencil when tested as specified in 4.3.4.

3.6 Type II ink. The type II ink shall have a water content of between 5 and 25 percent when tested as specified in 4.3.1. The diameter of a spot created by three drops of ink shall increase no more than 50 percent when tested as specified in 4.3.8.

3.7 Type IV ink. Type IV, class 1 ink shall be miscible with distilled water when tested as specified in 4.3.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of certification (see 3.2).

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The lot shall be all ink of one type and class from an identifiable batch offered for inspection at one time. The sample unit shall be 3 pounds of ink.

4.2.2.1 Visual examination. Examination of one filled unit container and the ink shall be made for defects listed in table II. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 defects expressed in terms of defects per hundred units.

TABLE II. Visual defects

Examine	Defects
Container	Not size, type, or class specified
	Seal not as required
	Interior and exterior coating, as applicable, not as specified
	Seepage or leakage from the container
	Closure not as specified
	Dented or distorted unit container
	Not marked as specified
Ink	Color not as specified
	Not homogeneous
	Presence of foreign matter

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4.2.2.2 Testing of the end item. The ink shall be tested as specified in 4.3. The inspection level shall be level I. Any nonconformance shall be cause for rejection of the batch.

4.2.2.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking are in accordance with the appendix of PPP-C-96.

4.3 Tests.

4.3.1 Physical tests. Specific gravity shall be tested as specified in Method 4183 of Fed. Test Method Std. No. 141. The water content of the type II ink shall be tested as specified in Method 4081. Nonconformance with table I or 3.6, as applicable shall constitute failure of this test.

4.3.2 Viscosity. Viscosity shall be measured using a Stormer Viscosimeter which has been previously standardized. Type II ink shall be tested using method b stated below. Types III and IV inks shall be tested using method a. Nonconformance with table I shall constitute failure of this test.

a. The Stormer Viscometer is set up and readied for use according to manufacturers recommendations.

b. The cup without the center baffle or thermometer holder is filled to 1/4 inch above the side vanes with Standard Oil S-600.

c. Place the test cup in the water bath and raise the platform until the oil covers the rotor to a depth of 1/4 inch. Set the stop knuckle so all future measurements will be taken in this same position. At this stage, the cup will be nearly full.

d. Insert a thermometer in the water bath and a thermometer in the test cup.

e. Adjust the temperature of the water bath so the test sample temperature will be 25° $\pm$ 0.1°C. The sample should be agitated during this period to make certain that the entire sample attains the proper temperature.

f. Remove the thermometer from the test cup.

g. Attach sufficient weight to the pully line to give 9.553 seconds per poise per 100 revolutions of the rotor. The spindle should be rotating at the time the count is started. (S-600 oil = 131.7 secs. @25°C.).

h. The weight required may vary, but is approximately 200 grams. Considerable departure from this, or an abrupt change in the weight required, may indicate a defective instrument.

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i. The procedure shall be repeated if the instrument has been adjusted or if 30 days has elapsed since the last standardization.

4.3.2.1 Method A.

a. The sample type III or type IV ink should be gently shaken or stirred by hand before pouring into the test cup.

b. Excess air bubbles in the sample can cause erroneous readings to result. Measurement should be avoided until the majority of the bubbles have cleared from the test sample.

c. The cup with the center baffle and thermometer holder (figure 1), is filled to 1/4 inch above the side vanes with the material to be measured.

d. Place the test cup in the water bath and raise the platform until the ink covers the rotor to a depth of 1/4 inch. At this stage the cup will be nearly full.

e. Insert a thermometer in the water bath and also in the test cup.

f. Carefully adjust the temperature of the water bath so the test sample temperature will be 20° $\pm$ 0.1°C. The sample should be agitated during this period to be certain that the entire sample attains the proper temperature.

g. The thermometer is not removed from the test cup.

h. The brake is released and the drive weight allowed to fall free of any obstruction. The spindle should be rotating when the stopwatch is started.

i. The number of seconds required for 100 revolutions of the spindle is measured and recorded as Stormer Viscosity.

j. The average of 5 readings is recorded as the viscosity of the material.

4.3.2.2 Method B.

a. The sample type II ink should be gently shaken or stirred by hand before pouring into the test cup.

b. Excess air bubbles in the sample can cause erroneous readings to result. Measurement should be avoided until the majority of the bubbles have cleared from the test sample.

c. The cup without the center baffle or thermometer holder (figure 2), is filled to 1/4 inch above the side vanes with sample material to be measured.

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d. Place the test cup in the water bath and raise the platform until the ink covers the rotor to a depth of 1/4 inch. At this stage the cup will be nearly full.

e. Insert a thermometer in the water bath and also in the test cup.

f. Carefully adjust the temperature of the water bath so the test sample temperature will be 25° ± 0.1°C. The sample should be agitated during this period to be certain that the entire sample attains the proper temperature.

g. The thermometer is then removed from the test cup.

h. The brake is released and the drive weight allowed to fall free of any obstruction. The spindle should be rotating when the stopwatch is started.

i. The number of seconds required for 100 revolutions of the spindle is measured and recorded as Stormer viscosity.

j. The average of 5 readings is recorded as the viscosity of the material.

4.3.3 Reproduction test. Two stencils conforming to UU-S-1205 shall be prepared on a commercial electric typewriter fitted with elite type. The copy shall be 30 lines by 6 inches across the stencil. Copy shall begin in the upper left corner of the stencil and a equal area of copy shall terminate in the lower right corner. Each area of copy shall be cut using all upper and lower case letters, numerals and symbols. Characters shall not penetrate the stencil. The type II or type III ink shall fill the cylinder of all electric motor driven closed cylinder stencil duplicating machine. When applicable, the cylinder shall be half filled with type IV ink. The following white papers shall be used:

Type II ink	Type III paper of UU-P-388
Type III ink	Type V paper of UU-P-121
Type IV ink	Type V paper of UU-P-121

The number of copies specified in 3.5 shall be run.

4.3.3.1 Fading test. Three copies shall be selected at random from the first 100 copies ahdl shall be tested as specified in Method 5660 of Fed. Test Method Std. No. 191. Nonconformance to 3.5 shall constitute failure of the test.

4.3.3.2 Drying time. After 1000 copies have been run, 1 copy shall be removed as the sheets are delivered from the machine and the drying time observed. Nonconformance to table I shall constitute failure of this test. The test shall be repeated after 2000 copies have been run and this copy shall be used for the test in 4.3.3.3.

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4.3.3.3 Smudge test. When type III ink is being tested, the copy shall be conditioned by dipping in water. When type II ink and type IV inks are being tested, the copy shall not be conditioned. The copy shall be placed on a desk top with printed side up and covered with a clean sheet of reproduction paper (see 4.3.3). Place a 100 gram, 3/4 inch diameter weight on the clean paper and draw across the printed copy. Evidence of smudge constitutes failure of this test.

4.3.3.4 Offset and strike through. The first and last 100 copies shall be examined after 24 hours. Evidence of offset, strike through or vehicle stain constitutes failure of this test.

4.3.4 Damage to stencil. Remove the stencil from the cylinder (4.5), place it between the paper specified in 4.3.3 and blot off the ink. Stencils used in testing type II ink and class 1, type IV ink shall be washed with water and blotted dry with absorbent paper. Open file folder on a flat surface, place the stencil on one side of the folder, inked side up, close the folder and rub the top to absorb ink from the stencil. Open the folder, remove stencil and return it to the folder, inked side down. Close the folder and file. After 24 hours examine the stencil. Tackiness or fill in constitutes failure of this test.

4.3.5 Leakage test. After completing the performance test in 4.3.3 check ink level, add sufficient ink to the cylinder to bring to the specified level and then remove the stencil. Wipe off any excess ink at the ink pad edges or around the clamps of the cylinder. If leakage takes place, examine the cause; the performance test shall be repeated with a new pad. Any leakage which is due to the ink should be reported. Place a protective cover over the ink pad and lock in place. Seal the cover to the pad by running several sheets of paper through the machine. Remove the cylinder from the machine and place it on a flat surface covered with several sheets of clean paper with the stop sign up and block to prevent rolling. After standing for 72 hours, examine the cylinder for leakage. With type IV ink a protective cover shall be placed over the ink pad, the feelers blocked so that the impression roller contacts the cylinder, and the cylinder rotated for 200 revolutions.

4.3.6 Separation. After completing the test specified in 4.3.5, remove the protective cover, place the unused stencil on the cylinder, and run 2,000 copies on the paper specified in 4.3.3 and examine. Oily blotches on the first 100 copies or broken copy on the last 100 copies constitutes failure of this test. Remove the new stencil and replace the stencil used in test specified in 4.3.3, and run 100 copies. Compare these copies with the last 100 copies run in the test specified in 4.3.3. Evidence of stencil deterioration constitutes failure of this test.

4.3.7 Accelerated stability test. Pour 10 grams of the sample ink into a 1/2-pint friction top can cover, and place into a controlled temperature atmosphere at 140°F for 48 hours. Skinning or caking shall constitute failure of this test. Failure of separated oil to remix the ink with agitation shall constitute failure of this test.

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4.3.8 Rate of spread. Stir sample for one minute with a hand paddle. Transfer sample to a pipetie. Release three drops, one on top of the other, from a height not to exceed 1 inch, onto a No. 2 Whatman filter paper placed on a glass substrate Measure the diameter of the black portion of the drop after 1 minute and after 5 minutes. An increase in diameter of more than 50 percent constitutes failure of this test.

4.3.9 Miscibility with distilled water (type IV, class 1). Mix throughly in a 50 ml beaker, one volume of type IV, class 1 ink with one volume of distilled water. Thickening, or appearance of curds which do not disappear upon mixing shall constitute failure of this test. Drop 0.5 to 1 ml of this mixture near the upper end of a clean glass plate 6 inches long and incline the plate 45°. Drop an equal amount of undiluted ink on a similarly inclined, clean, glass plate. After 4 hours, at room temperature, examine the ink-distilled-water mixture. Evidence of separation or thickening which retards the flow of the ink down the plate, as compared to ink not mixed with distilled water, constitutes failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Unit containers. Ink shall be packaged in quantities of one pound in containers conforming to PPP-C-96, type V, class 4 or 5. A tight-fitting metal inner seal shall be sealed in the neck of the container prior to closure. Exterior coating plan A shall be used.

5.2 Packaging, packing, and marking of unit containers. Unit containers specified in 5.1 shall be packaged, packed, and marked as specified for level A, B, or C, as specified in the appendix to PPP-C-96.

5.3 Special marking. Unit containers shall be marked with the following information:

"Use with A. B. Dick models 455, 438, 453, 450, 445, 437, 436, 435, 434, 432, 420, 418, 416, 412, 410 and 550 and similar machines equipped with fluid ink cylinders." "DO NOT USE IN PASTE CYLINDERS."

6. NOTES

6.1 Intended use. The inks covered by this specification are intended for use in single cylinder internally liquid inked cylinder stencil duplicating machines.

Type II inks are all-purpose inks which dry rapidly with a minimum of offset or strike-through on mimeograph papers and will give impressions with a minimum of offset on hard-sized papers. For use in open or closed single-cylinder duplicating machines.

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Type III inks give clean, clear impressions on hard-sized papers, smudge-proof when dry, and are not affected by water when dry. For use in closed single-cylinder duplicating machines only.

Type IV inks dry on contact with the paper. They operate well on hard-sized papers. For used in closed single-cylinder duplicating machines only. Type IV, class 1 water-base ink should be used in corrosion-resisting cylinders. Inked pads should be immediately covered after use to prevent evaporation and drying out of ink.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of ink required (see 1.2).

6.3 Supersession data. Type I ink of TT-I-564B has been deleted. Type II ink has been found to be a suitable replacement.

Custodians:

Army - GL
Air Force - 84

Review activity:

Army - ME

User activities:

Navy - MC, SA, YD

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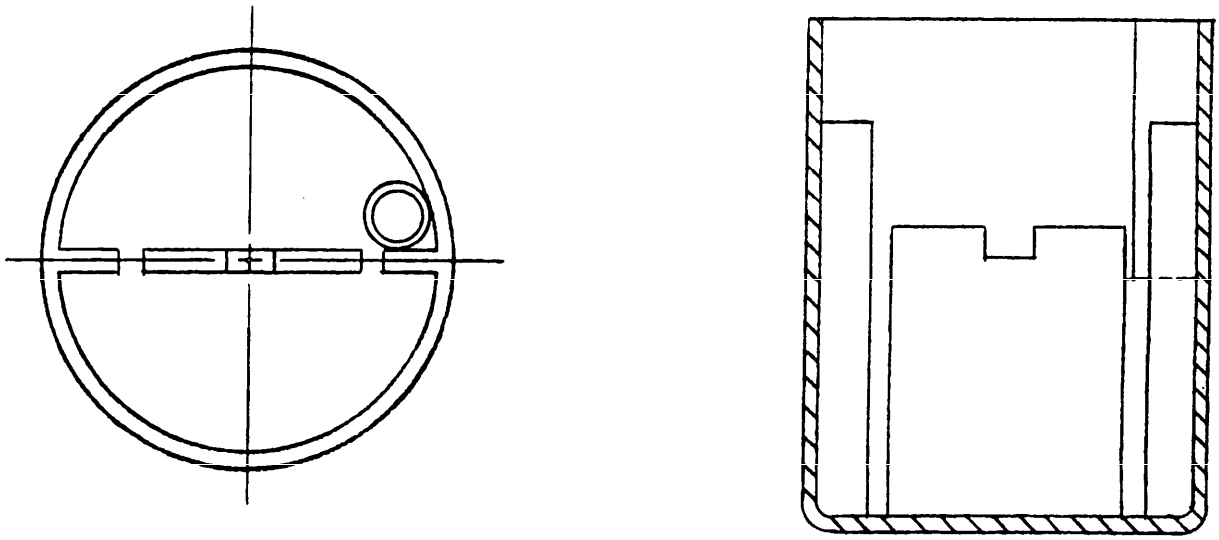


FIG. 1 TEST CUP WITH BAFFLE AND THERMOMETER HOLDER

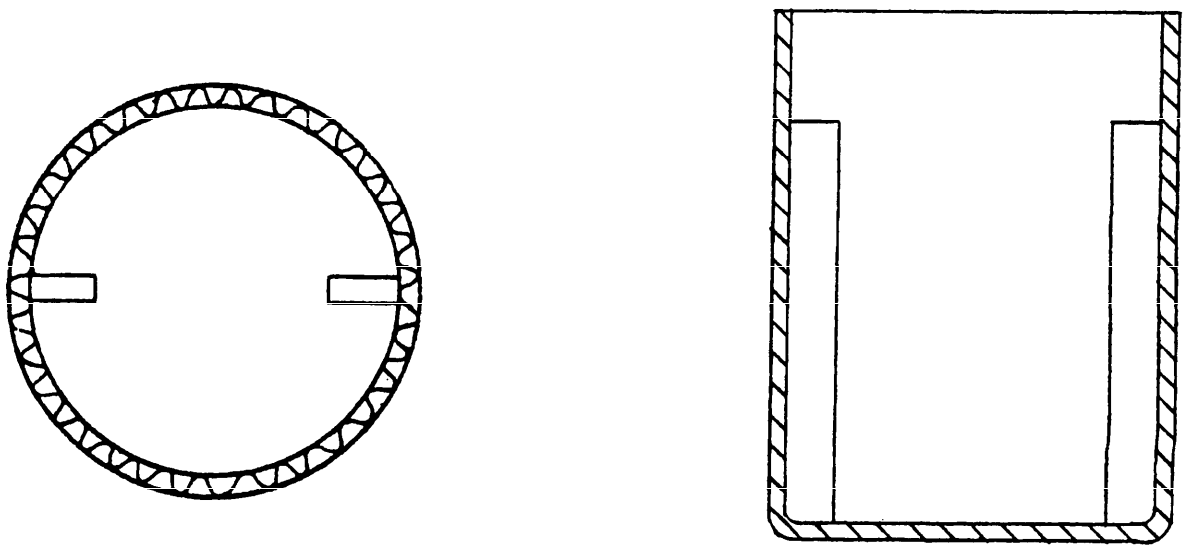


FIG. 2 TEST CUP WITHOUT BAFFLE

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DOCUMENT IDENTIFIER AND TITLE

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	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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